

Vacuum Native Selection Among Residual Mixtures on the 8×8 Torus

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Abstract

The WIN Paradigm derives the vacuum substrate as an 8×8 torus whose Laplacian spectrum has a 14 fold degeneracy at the mirror fixed point $\lambda = 4$. The 14 modes at this eigenvalue are the Standard Model particle content. Prior work derives the substrate geometry, the gauge structure, the mode classification, the entropy quantization (6.0, 5.5, 5.0) bits, the support classes $64, 48, 32 = N, V, N/2$, the $5 + 9$ split, and the $N/4$ values.

This paper addresses the selection problem: *when the vacuum presents a residual mixture of these modes, which mixture is preferred?* We analyze 300 samples classified into four residual mixture classes: $O_0 + O_1 + O_2$, $O_1 + O_2$, $O_0 + O_1$, and O_1 .

Three results are established. First, chemical potential variance is degenerate: all four classes have $V_{\text{chem}} = 0$. This functional cannot select. Second, O_1 is the unique Pareto optimal class on the two non-degenerate vacuum-native functionals (site entropy and participation ratio). It has the highest entropy (5.4700), the highest participation ratio (36.669), and a normalized selection score of $S = 2.000$, with all other classes ≤ 0.354 . Third, the selection is robust: a sensitivity sweep over within-class noise from $\sigma_H = 0$ to 0.20 and $\sigma_{\text{PR}} = 0$ to 2.0 gives $P(O_1 \in \text{Pareto}) \geq 0.986$ across the entire sweep.

The bridge to the 8×8 torus is the independence decomposition. The four mixture frequencies are consistent with three independent binary survivals: O_0 with probability $p_0 \approx 0.553$, O_1 with probability $p_1 = 1.000$, and O_2 with probability $p_2 \approx 0.933$. O_1 is always present; O_0 and O_2 are optional. The chi-squared test for the independence model gives $\chi^2 = 0.81$ with 1 degree of freedom ($p \approx 0.37$): the model is not rejected. The selection principle is structural, not statistical. It is anti-frequency: the rarest configuration is the vacuum-native preferred working state.

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1 Introduction

1.1 The framework

The WIN Paradigm identifies the vacuum substrate with the 8×8 torus. The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue pairs with its mirror at equal multiplicity, except $\lambda = 4$, which is the unique self-paired eigenvalue with multiplicity 14.

The 14 modes at $\lambda = 4$ are the Standard Model particle content. Their support values are exactly 64, 48, $32 = N, V, N/2$. Their entropies are quantized to 6.0, 5.5, 5.0 bits. Their charges split into $+2/3$ (outer mesh, 6 modes) and $-5/3$ (middle mesh, 8 modes), totaling -1 .

1.2 The problem

Given this substrate, a *residual mixture* is a subset of the three structural components O_0, O_1, O_2 that survives a cancellation or filtering process. The question is: *which mixture is preferred?*

The empirical data comprise 300 samples, each classified into one of four residual mixture classes. The observed frequencies are shown in Table 1.

Table 1: Residual mixture frequencies.

Class	Frequency
$O_0 + O_1 + O_2$	153/300
$O_1 + O_2$	127/300
$O_0 + O_1$	13/300
O_1	7/300

For each class, three vacuum native functionals are computed: site entropy H , participation ratio PR, and chemical potential variance V_{chem} .

1.3 What this paper establishes

1. Chemical-potential variance is degenerate and cannot select.
2. O_1 is the unique Pareto-optimal residual mixture on the non-degenerate functionals.
3. The selection is robust under sampling noise.
4. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$.
5. The selection is structural: O_1 is the minimal surviving configuration, and the selection is anti frequency.

1.4 The bridge

The three components O_0, O_1, O_2 are identified with the three support classes of the 14-mode multiplet:

$$\begin{aligned}
 O_0 &\leftrightarrow \text{support 64 (2 modes: } \nu, e), \\
 O_1 &\leftrightarrow \text{support 48 (8 modes: } u, d, s, \mu, \text{dark}, c, \tau, b), \\
 O_2 &\leftrightarrow \text{support 32 (4 modes: } W, Z, H, t).
 \end{aligned}$$

This identification is proposed as a hypothesis. It is consistent with the independence decomposition and with the observed selection of O_1 .

2 The 8×8 Torus and the 14-Mode Multiplet

2.1 The substrate

The 8×8 torus has 64 sites with periodic boundary conditions and coordination number 4. Its Laplacian has the eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad (1)$$

with $(n_1, n_2) \in \{0, 1, \dots, 7\}^2$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ with 13 distinct eigenvalues. Table 2 lists the full spectrum.

Table 2: Laplacian spectrum of the 8×8 torus.

λ	Multiplicity
0.0000	1
0.5858	4
1.1716	4
2.0000	4
2.5858	8
3.4142	4
4.0000	14
4.5858	4
5.4142	8
6.0000	4
6.8284	4
7.4142	4
8.0000	1

The 50 paired modes cancel under the mirror involution. The 14 modes at $\lambda = 4$ are self-paired and survive.

2.2 The 14-mode multiplet

The 14 modes split by support into three classes (Table 3).

Table 3: Support classes of the 14-mode multiplet.

Support	Framework constant	Modes	Charge
64	N	2 (ν, e)	0, -1
48	V	8 ($u, d, s, \mu, \text{dark}, c, \tau, b$)	$+2/3, -1/3, -1$
32	$N/2$	4 (W, Z, H, t)	$+1, 0, 0, +2/3$

Per-mode entropies are quantized: 6.0 bits (support 64), 5.5 bits (support 48), 5.0 bits (support 32). The quantization spacing is exactly 0.5 bits.

The two-mesh split assigns the 6 modes of support $64 \cup 32$ to the outer mesh (charge $+2/3$) and the 8 modes of support 48 to the middle mesh (charge $-5/3$), totaling -1 .

2.3 The three structural components

The three support classes are the natural structural components of the multiplet:

- O_0 = support 64 (the core, 2 modes),
- O_1 = support 48 (the middle mesh, 8 modes),
- O_2 = support 32 (the outer mesh, 4 modes).

A residual mixture is a subset of $\{O_0, O_1, O_2\}$.

3 Residual Mixtures

3.1 Definition

A *residual mixture* is a subset of the three structural components that survives a cancellation or filtering process in a given vacuum sample. The four observed mixtures are listed in Table 4.

Table 4: Residual mixtures and their mode counts.

Mixture	Components	Size (modes)
$O_0 + O_1 + O_2$	all three	14
$O_1 + O_2$	middle + outer	12
$O_0 + O_1$	core + middle	10
O_1	middle only	8

3.2 Frequency data

Across 300 samples, the frequencies are those in Table 1. Two structural facts are immediately visible:

1. O_1 is present in every observed class. There is no O_0 -only, O_2 -only, or $O_0 + O_2$ sample.
2. O_0 and O_2 are optional additions. The four observed mixtures are exactly $\{O_0 \text{ or not}\} \times \{O_2 \text{ or not}\}$ with O_1 always present.

3.3 Vacuum-native functionals

For each mixture class, three functionals are computed:

- **Site entropy** H : Shannon entropy of the mode population.
- **Participation ratio** PR: inverse participation of the mode vector.
- **Chemical-potential variance** V_{chem} : variance of the mean-field chemical potential.

Mean values by class are given in Table 5.

Table 5: Mean vacuum-native functionals by mixture class.

Class	Entropy	PR	V_{chem}
$O_0 + O_1 + O_2$	5.4295	34.631	0.000000
$O_1 + O_2$	5.4395	34.848	0.000000
$O_0 + O_1$	5.4361	34.868	0.000000
O_1	5.4700	36.669	0.000000

4 Deterministic Selection

4.1 Degeneracy of chemical-potential variance

All four classes have $V_{\text{chem}} = 0.000000$. The functional is degenerate and cannot select. Any apparent ordering by V_{chem} is a tie and must be discarded. The effective selection functionals reduce to

$$F_{\text{eff}}(c) = (H(c), \text{PR}(c)). \quad (2)$$

4.2 Pareto dominance

Definition 1. A class c is Pareto-optimal if no other class d dominates it, i.e., no d has $H(d) \geq H(c)$ and $\text{PR}(d) \geq \text{PR}(c)$ with strict inequality in at least one functional.

Applying this to Table 5:

- O_1 has the highest entropy (5.4700) and the highest PR (36.669). It dominates every other class.
- No other class dominates O_1 , because no other class has both higher H and higher PR.

Proposition 1. The Pareto set is $\{O_1\}$.

4.3 Borda rank aggregation

Entropy ranking (highest to lowest): $O_1, O_1 + O_2, O_0 + O_1, O_0 + O_1 + O_2$.

PR ranking (highest to lowest): $O_1, O_0 + O_1, O_1 + O_2, O_0 + O_1 + O_2$.

Borda scores (sum of ranks, lower is better) are given in Table 6.

Table 6: Borda rank aggregation.

Class	Entropy rank	PR rank	Borda
O_1	1	1	2
$O_0 + O_1$	3	2	5
$O_1 + O_2$	2	3	5
$O_0 + O_1 + O_2$	4	4	8

Proposition 2. The Borda winner is O_1 .

4.4 Normalized selection score

Define the normalized selection score

$$S(c) = \frac{H(c) - H_{\min}}{H_{\max} - H_{\min}} + \frac{\text{PR}(c) - \text{PR}_{\min}}{\text{PR}_{\max} - \text{PR}_{\min}}. \quad (3)$$

The values are given in Table 7.

Proposition 3. $\arg \max_c S(c) = O_1$.

4.5 Result

Theorem 1 (Deterministic selection). O_1 is the unique vacuum-native selected residual mixture on the non-degenerate functionals. The selection is anti-frequency: the rarest mixture (7/300) is chosen over the majority (153/300).

Table 7: Normalized selection scores.

Class	S
O_1	2.000
$O_1 + O_2$	0.353
$O_0 + O_1$	0.279
$O_0 + O_1 + O_2$	0.000

5 Robustness Under Sampling Noise

5.1 The concern

O_1 has $n = 7$. The sample mean may not reflect the true mean. A selection that survives only at zero noise is not reliable.

5.2 Sensitivity sweep

We resample the class means with within-class standard deviations σ_H and σ_{PR} , compute the Borda winner and the Pareto set, and record the frequency of O_1 selection. Sample sizes n per class are held at their observed values (153, 127, 13, 7). Four thousand Monte Carlo trials are performed per cell. Selected results are shown in Table 8.

Table 8: Sensitivity sweep (selected rows).

σ_H	σ_{PR}	$P(O_1 \text{ win})$	$P(O_1 \in \text{Pareto})$
0.000	0.00	1.0000	1.0000
0.010	1.00	1.0000	1.0000
0.020	2.00	0.9855	1.0000
0.050	2.00	0.9173	0.9988
0.100	0.50	0.7630	1.0000
0.200	0.50	0.6158	1.0000

5.3 Interpretation

Two regimes are visible:

- **Pareto robustness.** Across the entire sweep, $P(O_1 \in \text{Pareto}) \geq 0.986$. The Pareto selection is essentially bulletproof: no plausible noise level makes O_1 dominated.
- **Borda robustness.** $P(O_1 \text{ wins}) \geq 0.92$ for $\sigma_H \leq 0.05$, degrading to ≈ 0.61 at $\sigma_H = 0.20$. The Borda win is carried by the participation-ratio channel.

5.4 Margin analysis

$$\text{Entropy gap : } O_1 - \max(\text{other}) = +0.0305, \quad (4)$$

$$\text{PR gap : } O_1 - \max(\text{other}) = +1.8010, \quad (5)$$

$$\text{SE}_H(O_1) \text{ at } \sigma_H = 0.05 : 0.0189 \quad (62\% \text{ of entropy gap}), \quad (6)$$

$$\text{SE}_{PR}(O_1) \text{ at } \sigma_{PR} = 0.5 : 0.1890 \quad (10.5\% \text{ of PR gap}). \quad (7)$$

The entropy channel is confirmatory. The PR channel is decisive.

5.5 Result

Theorem 2 (Robustness). O_1 is the unique Pareto-optimal residual mixture across the entire noise sweep. The Borda result is robust for $\sigma_H \leq 0.05$. The selection is anchored by participation ratio.

6 Independence Decomposition of Mixture Frequencies

6.1 The observation

O_1 appears in all four classes. O_0 and O_2 appear only in combination with O_1 . This suggests a binary survival model.

6.2 The model

Let p_0, p_1, p_2 be the survival probabilities of the three components. Under independence:

$$P(O_0 + O_1 + O_2) = p_0 p_1 p_2, \quad (8)$$

$$P(O_1 + O_2) = (1 - p_0) p_1 p_2, \quad (9)$$

$$P(O_0 + O_1) = p_0 p_1 (1 - p_2), \quad (10)$$

$$P(O_1) = (1 - p_0) p_1 (1 - p_2). \quad (11)$$

6.3 Parameter extraction

From the data:

$$p_1 = P(O_1 \text{ present}) = 1.000, \quad (12)$$

$$p_0 = P(O_0 \text{ present}) = \frac{153 + 13}{300} = \frac{166}{300} = 0.5533, \quad (13)$$

$$p_2 = P(O_2 \text{ present}) = \frac{153 + 127}{300} = \frac{280}{300} = 0.9333. \quad (14)$$

6.4 Predictions vs. observations

Table 9 compares predictions to observations.

Table 9: Independence model: predicted vs. observed frequencies.

Mixture	Predicted	Observed	Difference
$O_0 + O_1 + O_2$	0.5164	0.5100	-0.0064
$O_1 + O_2$	0.4169	0.4233	+0.0064
$O_0 + O_1$	0.0369	0.0433	+0.0064
O_1	0.0298	0.0233	-0.0065

6.5 Chi-squared test

Expected counts ($\times 300$): 154.92, 125.07, 11.07, 8.94. Observed counts: 153, 127, 13, 7.

$$\chi^2 = \frac{(153 - 154.92)^2}{154.92} + \frac{(127 - 125.07)^2}{125.07} + \frac{(13 - 11.07)^2}{11.07} + \frac{(7 - 8.94)^2}{8.94} = 0.811. \quad (15)$$

Degrees of freedom = 1, $p \approx 0.37$.

Proposition 4. *The independence model is not rejected.*

6.6 Structural reading

The mixture frequencies are consistent with three independent binary survivals (Table 10).

Table 10: Survival probabilities of the three structural components.

Component	Survival probability
O_0	$p_0 \approx 0.553$
O_1	$p_1 = 1.000$
O_2	$p_2 \approx 0.933$

O_1 is topologically or symmetry-protected: it survives in every sample. O_0 and O_2 are optional. The four observed mixtures are exactly the four combinations $\{O_0\} \times \{O_2\}$ with O_1 always present.

6.7 Result

Theorem 3 (Independence decomposition). *The residual mixture frequencies decompose as independent binary survivals with O_1 always present. The selected class O_1 is the minimal surviving configuration.*

7 Bridge to the 8×8 Torus Geometry

7.1 The identification hypothesis

The three structural components O_0, O_1, O_2 are proposed to correspond to the three support classes of the 14-mode multiplet (Table 11).

Table 11: Proposed bridge: components to support classes.

Component	Support	Modes	Framework constant
O_0	64	2 (ν, e)	N
O_1	48	8 ($u, d, s, \mu, \text{dark}, c, \tau, b$)	V
O_2	32	4 (W, Z, H, t)	$N/2$

7.2 Consistency with observed frequencies

The independence decomposition gives $p_1 = 1.000$. If O_1 is the support-48 middle mesh, then the middle mesh is topologically protected. This is consistent with the framework's two-mesh structure:

- The middle mesh (support 48) carries charge $-5/3$ and is the largest single class (8 of 14 modes).
- The outer mesh (support $64 \cup 32$) carries charge $+2/3$.

The middle mesh being always present means the vacuum never presents a configuration without the 8-mode core.

7.3 Consistency with selection

The selected class $O_1 = \text{support } 48$ is the largest single support class (8 modes). Its mean per-mode entropy is 5.5 bits, intermediate between the core (6.0) and the outer (5.0).

7.4 Prediction of the bridge hypothesis

Under the hypothesis, the vacuum-native functionals computed directly from the mode vectors of the support 48 class should be extremal relative to the other support classes and their unions. Specifically:

- The mean PR of the support 48 class should be higher than the mean PR of the support 64 and support-32 classes.
- The mean entropy of the support-48 class should be higher than the mean entropy of the full 14-mode multiplet.

7.5 Open question

The observed PR values in the mixture analysis (34.6 to 36.7) do not match the naive per-mode PR values of the 8×8 torus modes (which range over 32 to 64 by support class). This discrepancy must be resolved before the bridge is complete. Two possibilities:

1. The mixture PR is a collective or joint quantity, not a mean of per-mode PRs.
2. The mixture analysis uses a different definition of participation ratio.

The bridge is consistent with the independence decomposition and with the selection of O_1 , but the exact functional correspondence remains open.

7.6 Result

Proposition 5 (Bridge hypothesis). *The identification of O_0, O_1, O_2 with the three support classes of the 14-mode multiplet is consistent with all available data. It is proposed as the structural bridge between the vacuum-native selection result and the torus geometry.*

8 What Is Derived, Hypothesized, and Open

8.1 Derived

1. Chemical-potential variance is degenerate (all four classes = 0.000000).
2. O_1 is the unique Pareto-optimal class on the non degenerate functionals.
3. O_1 is the Borda winner with score 2, next best 5.
4. O_1 has the highest normalized selection score ($S = 2.000$).
5. $P(O_1 \in \text{Pareto}) \geq 0.986$ across the entire noise sweep ($\sigma_H \leq 0.20$, $\sigma_{\text{PR}} \leq 2.0$).
6. $P(O_1 \text{ wins Borda}) \geq 0.92$ for $\sigma_H \leq 0.05$.
7. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$.
8. The independence model is not rejected ($\chi^2 = 0.81$, $p \approx 0.37$).
9. The selection is anti-frequency: the rarest class is chosen.

8.2 Hypothesized

1. O_0 corresponds to support 64 (2 modes: ν, e).
2. O_1 corresponds to support 48 (8 modes: middle mesh).
3. O_2 corresponds to support 32 (4 modes: outer mesh).

8.3 Open

1. The exact definition of the mixture participation ratio.
2. The exact mechanism by which O_1 is topologically protected.
3. Whether the independence of O_0, O_1, O_2 is exact or approximate.
4. The full D_4 orbit decomposition of the 14 modes and its relation to the support classes.
5. Whether the bridge hypothesis can be tested with a direct spectral calculation.

9 Discussion

9.1 The selection principle

The vacuum native selection among residual mixtures is Pareto dominance on site entropy and participation ratio. The selected class is O_1 . The selection is anti-frequency: the rarest mixture is preferred.

9.2 The structural origin

The independence decomposition suggests that O_1 is not merely the best class statistically; it is the class that is always present. The other components, O_0 and O_2 , are optional additions. The vacuum-native selection does not choose O_1 from a set of equally plausible alternatives; it recognizes O_1 as the irreducible core of the residual mixture.

9.3 The reading in the framework's language

If the identification hypothesis is correct:

- O_0 (support 64) is the 2-mode core (ν, e).
- O_1 (support 48) is the 8-mode middle mesh ($u, d, s, \mu, \text{dark}, c, \tau, b$).
- O_2 (support 32) is the 4-mode outer mesh (W, Z, H, t).

The always-present O_1 is the middle mesh. The selection of O_1 alone corresponds to a vacuum configuration in which only the middle mesh survives.

The anti frequency character of the selection is now structural: the full 14-mode multiplet is the majority configuration (153/300), but it is not the extremal one. The extremal configuration is the minimal one, containing only the protected middle mesh.

9.4 Relation to the self-canceling vacuum

In the self canceling vacuum, 50 paired modes cancel and 14 self-paired modes survive at $\lambda = 4$. The present result suggests a further selection among the 14: the middle mesh (support 48) is the irreducible residual.

This is consistent with the framework’s two-mesh structure. The middle mesh carries charge $-5/3$, the outer mesh $+2/3$, totaling -1 . The selection of the middle mesh alone would leave a net charge of $-5/3$, which is not observed. The bridge therefore requires a separate charge-balancing mechanism, or the identification of O_1 with the middle mesh must be revised.

9.5 Caveat

The identification of O_0, O_1, O_2 with support classes is a hypothesis, not a derivation. The independence structure of the frequencies and the selection of O_1 are derived from the data. The geometric identification is proposed as the simplest bridge consistent with the data.

10 Conclusion

The vacuum-native selection among residual mixtures on the 8×8 torus is complete.

1. Chemical-potential variance is degenerate.
2. O_1 is the unique Pareto-optimal residual mixture on site entropy and participation ratio.
3. The selection is robust to sampling noise ($P(O_1 \in \text{Pareto}) \geq 0.986$ across the sweep).
4. The mixture frequencies decompose as independent binary survivals with $p_0 \approx 0.553$, $p_1 = 1.000$, $p_2 \approx 0.933$.
5. The selection is anti-frequency: the rarest mixture is chosen.
6. The three components O_0, O_1, O_2 are identified with the three support classes of the 14-mode multiplet (64, 48, 32).

The vacuum native preferred working state is O_1 .

A Numerical Values

B Reproducibility

All numerical results in this paper are reproducible from the summary statistics in Tables 1 and 5. The deterministic selection follows from direct application of Pareto dominance and Borda rank aggregation. The sensitivity sweep follows from Monte Carlo resampling of class means with the stated standard deviations and observed sample sizes. The independence decomposition follows from the frequency data by direct parameter extraction and a chi-squared test with one degree of freedom. The 8×8 torus spectrum in Table 2 follows from the Laplacian eigenvalues $\lambda(n_1, n_2) = 4 - 2 \cos(2\pi n_1/8) - 2 \cos(2\pi n_2/8)$ over $(n_1, n_2) \in \{0, \dots, 7\}^2$.

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Table 12: Summary of numerical values.

Quantity	Value
d	4
N	64
H	16
V	48
L	8
Modes at $\lambda = 4$	14
Support classes	64, 48, 32
Entropy quantization	6.0, 5.5, 5.0 bits
Total samples	300
$O_0 + O_1 + O_2$ frequency	153/300
$O_1 + O_2$ frequency	127/300
$O_0 + O_1$ frequency	13/300
O_1 frequency	7/300
O_1 entropy	5.4700
O_1 PR	36.669
O_1 Pareto probability (worst case)	0.986
O_1 Borda probability ($\sigma_H \leq 0.05$)	≥ 0.92
p_0	0.5533
p_1	1.0000
p_2	0.9333
χ^2 (independence)	0.811
p -value	0.37

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